USAFA / DEPARTMENT OF ASTRONAUTICS

ASTRO 331 | KESTRELSAT CAPSTONE

Preliminary Data Review • Instructor Improvement Notes

1. Engineering / design improvements (hardware, software, integration, iteration)

Solid troubleshooting and diagnostics runs.
Issues with a slew/pointing bias.

2. Experimental data and analysis (plots, predicted vs. measured, unexpected results)

Useful pointing error plots, keep for final and compare
Quantify settling time, error, wheel speed saturation

3. Requirements and verification (thresholds, additional tests, stretch objectives)

Requirements are verifiable
Complete remaining requirements.

4. Instructor support / project constraints (material delays, resources, issues to resolve)

Saturation → resister values → show/compare data
Show code & discuss.

Priority actions before the final presentation

Action / next experiment	Owner	Target date

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Preliminary Data Review • Rubric

Team / Cadets: Astro-Nots	Instructor / Reviewer: Maj Boyer
Project / Subsystem: ADCS	Date / Section: 23 Oct / MSA

CRITERION	EVALUATION FOCUS	MAX	SCORE
Data – quantity and relevance	At least two distinct, relevant plots; physical test data and theoretical/expected performance; avoid redundant figures.	10	16
Data – plot quality / readability	Readable titles, labeled axes and units, test conditions, legends, reference values, and identified data sources.	20	19
Data – engineering interpretation	Explain trends, predicted-versus-measured differences, uncertainty, failed tests, and implications for next steps.	10	8
Hardware / software engineering progress	Evidence of assembled/integrated hardware, working code, instrumentation, physical tests, and design iteration.	20	15
Requirements and verification	At least 3 measurable threshold and 1 stretch requirement; planned verification and demonstrated/in-progress status.	15	15
Plan to final presentation	Specific remaining tasks and tests, realistic sequence, schedule risks, mitigation, and verification by Lesson 31.	10	8
Innovation and technical ambition	Meaningful, feasible, testable capability expansion beyond baseline work without jeopardizing thresholds.	10	8
Professionalism and technical understanding	Clear team presentation, defensible design decisions and responses, and documented GenAI usage.	5	4
TOTAL SCORE		100	87

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